ACS APPLIED MATERIALS & INTERFACES

Research Article

Cite This: ACS Appl. Mater. Interfaces XXXX, XXX, XXX-XXX

www.acsami.org

Role of a Modulator in the Synthesis of Phase-Pure NU-1000

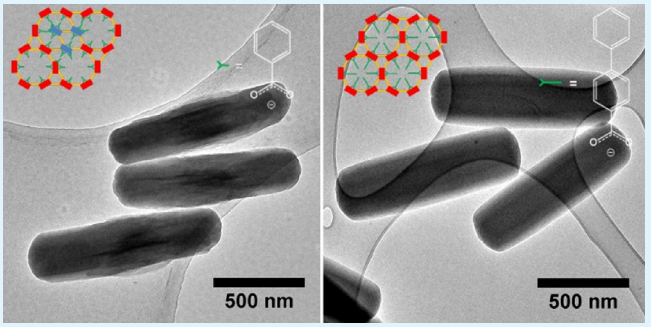
Thomas E. Webber,[†] Wei-Guang Liu,[†,‡] Sai Puneet Desai,[†] Connie C. Lu,[†] Donald G. Truhlar,[†,‡] and R. Lee Penn*[,†]

[†]Department of Chemistry, University of Minnesota, Minneapolis, Minnesota 55455-0431, United States

[‡]Chemical Theory Center and Minnesota Supercomputing Institute, University of Minnesota, Minneapolis, Minnesota 55455, United States

Supporting Information

ABSTRACT: NU-1000 is a robust, mesoporous metal−organic framework (MOF) with hexazirconium nodes ($[Zr_6O_{16}H_{16}]^{8+}$, referred to as oxo−Zr_6 nodes) that can be synthesized by combining a solution of $ZrOCl_2{\cdot}8H_2O$ and a benzoic acid modulator in *N*,*N*-dimethylformamide with a solution of linker (1,3,6,8-tetrakis(*p*-benzoic acid)pyrene, referred to as H_4TBAPy) and by aging at an elevated temperature. Typically, the resulting crystals are primarily composed of NU-1000 domains that crystallize with a more dense phase that shares structural similarity with NU-901, which is an MOF composed of the same linker molecules and nodes. Density differences between the two polymorphs arise from the differences in the node orientation: in NU-1000, the oxo−Zr_6 nodes rotate 120° from node to node, whereas in NU-901, all nodes are aligned in parallel. Considering this structural difference leads to the hypothesis that changing the modulator from benzoic acid to a larger and more rigid biphenyl-4-carboxylic acid might lead to a stronger steric interaction between the modulator coordinating on the oxo−Zr_6 node and misaligned nodes or linkers in the large pore and inhibit the growth of the more dense NU-901-like material, resulting in phase-pure NU-1000. Side-by-side reactions comparing the products of synthesis using benzoic acid or biphenyl-4-carboxylic acid as a modulator produce structurally heterogeneous crystals and phase-pure NU-1000 crystals. It can be concluded that the larger and more rigid biphenyl-4-carboxylate inhibits the incorporation of nodes with an alignment parallel to the neighboring nodes already residing in the crystal.



KEYWORDS: *metal−organic frameworks, metal infiltration, modulator, NU-1000, phase-pure synthesis*

INTRODUCTION

Metal−organic frameworks (MOFs) based on hexazirconium nodes with strong oxo−Zr bonds ($[Zr_6O_{16}H_{16}]^{8+}$, referred to as oxo−Zr_6 nodes) have exceptional thermal, chemical, and mechanical stability.[1−3] MOFs with oxo−Zr_6 nodes are structurally diverse.[4−13] Two polymorphs of an MOF composed of $Zr_6(\mu_3\text{-}OH)_4(\mu_3\text{-}O)_4(OH)_4(OH_2)_4$ nodes and the linker (1,3,6,8-tetrakis(*p*-benzoic acid)pyrene; H_4TBAPy), namely, NU-1000[14] and NU-901,[15] are shown in Figure 1.

NU-1000 is a highly porous MOF that has been used as a scaffold for anchoring a variety of metals[16−21] and bimetallic complexes.[22,23] The large hexagonal pores of NU-1000 are 31 Å in diameter, which means that a broad range of species (e.g., precursors for installation of active sites, reactants, and products of catalytic reactions) can readily diffuse into and out of the material.

NU-1000 is typically synthesized by combining a solution of $ZrOCl_2{\cdot}H_2O$ and benzoic acid in *N*,*N*-dimethylformamide (DMF) with a solution of TBAPy, also in DMF, and by aging at an elevated temperature.[24] The typical product crystals (Figure 2) are primarily composed of NU-1000 (0.486 g/cm^3), but they also contain a more dense phase that shares structural similarity with NU-901 (0.704 g/cm^3), which has a higher node density. The structural heterogeneity severely complicates the structural characterization of the anchored active species on the Zr_6-based nodes using methods such as X-ray diffraction, difference envelope density analysis, atomic structure imaging by transmission electron microscopy (TEM), and extended X-ray absorption spectroscopy fine structure. The structural heterogeneity is also undesirable from a materials perspective in that the more dense material contains only smaller pores, which means that larger species (e.g., bimetallic complexes or large substrates) cannot access the interior nodes, resulting in a lower usable mass per gram of the material synthesized.

NU-1000 and NU-901 (Figure 3) differ in node orientation: in NU-1000, the oxo−Zr_6 nodes rotate 120° from node to node, whereas in NU-901, all nodes are aligned in parallel. It was proposed that in the ordinary NU-1000 synthesis, the large pores of NU-1000 are partially occupied by oxo−Zr_6 nodes, which results in the formation of the NU-901-like structure.

Received: August 2, 2017
Accepted: October 17, 2017

ACS Publications © XXXX American Chemical Society

DOI: 10.1021/acsami.7b11348
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

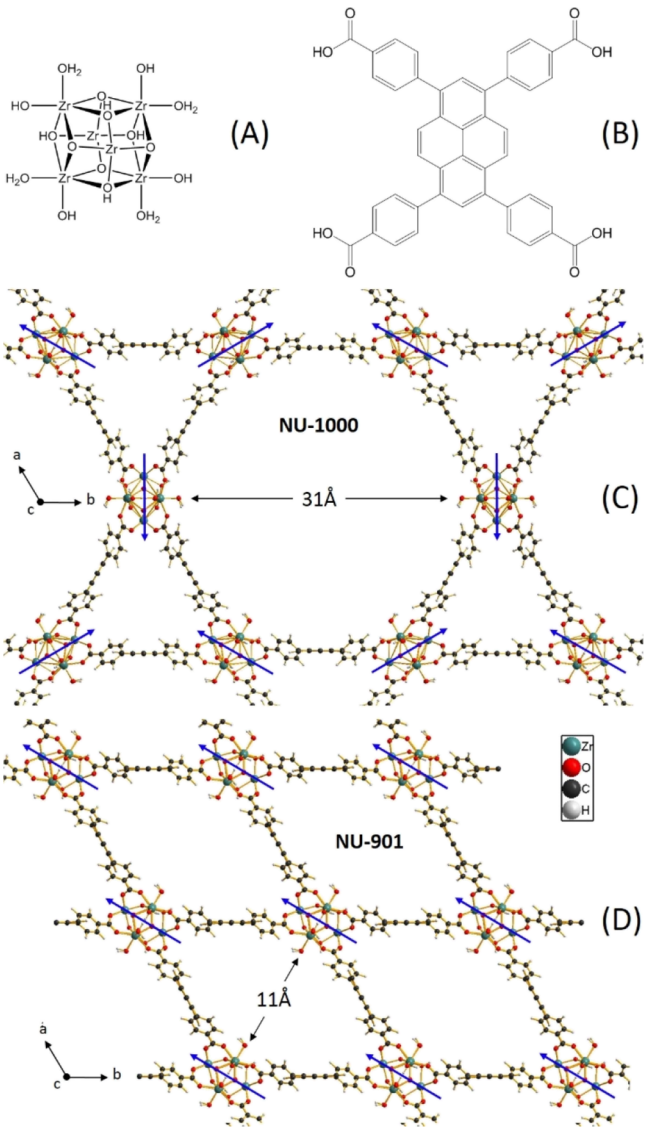


Figure 1. Structure of the (A) oxo−Zr_6 node, (B) linker molecule, (C) NU-1000, and (D) NU-901. Blue arrows highlight the orientations of the nodes within the NU-1000 and NU-901 structures.

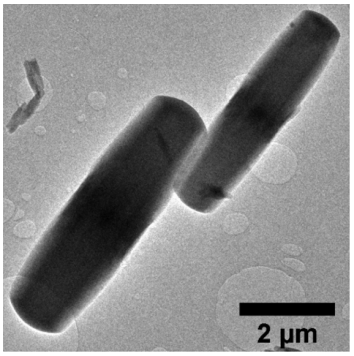


Figure 2. TEM image of NU-1000 crystals synthesized using benzoic acid as the modulator. The heterogeneous contrast observed is consistent with the presence of a denser material in the center regions of the crystals.

NU-1000 is similar to other MOFs with oxo−Zr_6 nodes in that acid modulators are added to promote the MOF crystal growth by slowing the reaction kinetics, thereby facilitating the formation of the nodes and preventing the fast precipitation of disordered and/or amorphous materials.[25−27] In the solution of $ZrOCl_2·H_2O$ and the modulator, the zirconium atoms are ligated mainly to the modulator. Combining this solution with the solution containing the linker results in the replacement of modulator molecules with linkers, and in this way, the MOF is synthesized.

The growth process of NU-1000 has been discussed in detail elsewhere[28] and is briefly discussed here. In the presence of excess benzoic acid as the modulator, the Zr_6 nodes are mostly capped by benzoates, including those nodes on the crystal surface that are dangling and thermally allowed to rotate (the calculated barrier is 8.6 kcal/mol) about the C−C bonds between pyrene and benzoates. To grow the crystal, TBAPy has to replace the benzoate modulators to connect the Zr_6 nodes. When two Zr_6 nodes are connected at a 120° angle, NU-1000 is formed. When two Zr_6 nodes are connected in parallel, the more dense, NU-901-like structure is formed.

The above structural considerations lead to a hypothesis that phase-pure NU-1000 could be synthesized by replacing the benzoic acid modulator by a larger and more rigid biphenyl-4-carboxylic acid.

RESULTS AND DISCUSSION

This synthetic condition change was inspired by the fact that the NU-901-like phase has higher node density. If a long and rigid modulator, such as biphenyl-4-carboxylic acid, is used in the synthesis of NU-1000, it could cause a stronger steric interaction between the modulator coordinating on the Zr_6 node and the misaligned nodes or linkers in the large pore, and this could interrupt the formation of the NU-901-like phase and improve the phase purity of NU-1000, as shown in the bottom of Figure 3. The choice of the modulators may also affect the growth rate of NU-1000 because the organic linker needs to replace the modulator coordinated on the Zr_6 node to grow the crystal, and Zr_6 nodes capped with different modulators have different dissociation energies. We calculated the reaction energies and reaction-free energies to exchange the modulators coordinating on the Zr_6 node with water, as shown in Scheme 1, and the results are shown in Table 1. The higher exchange energy of biphenyl-4-carboxylic acid indicates that it bonds more strongly to the Zr_6 node, and this may slow down the crystal growth and decrease the defect density in NU-1000.

Indeed, a replacement of benzoic acid with biphenyl carboxylic acid results in the production of phase-pure NU-1000. Side-by-side reactions comparing the products of synthesis using benzoic acid or biphenyl-4-carboxylic acid as the modulator produce structurally heterogeneous crystals and phase-pure NU-1000 crystals. The synthesis employed is a modified version of the procedure developed by Wang et al.,[24] and the detailed procedure can be found in the Supporting Information. All experimental variables, with the exception of the choice of the modulator, were kept constant. X-ray diffraction (Figure S1, where figures with an S prefix are in the Supporting Information) and N_2 sorption (Figure S2) results demonstrate phase purity and excellent crystallinity and porosity. However, it is the results from imaging via TEM that provide the most compelling evidence that the product crystals are structurally homogeneous (Figure 4). The particles exhibit uniform contrast from tip to tip and ∼2.6 nm lattice spacings, consistent with the NU-1000, that span the entire particle. No variations in contrast consistent with the denser material or with strain were observed. Thus, we conclude that the product

DOI: 10.1021/acsami.7b11348
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

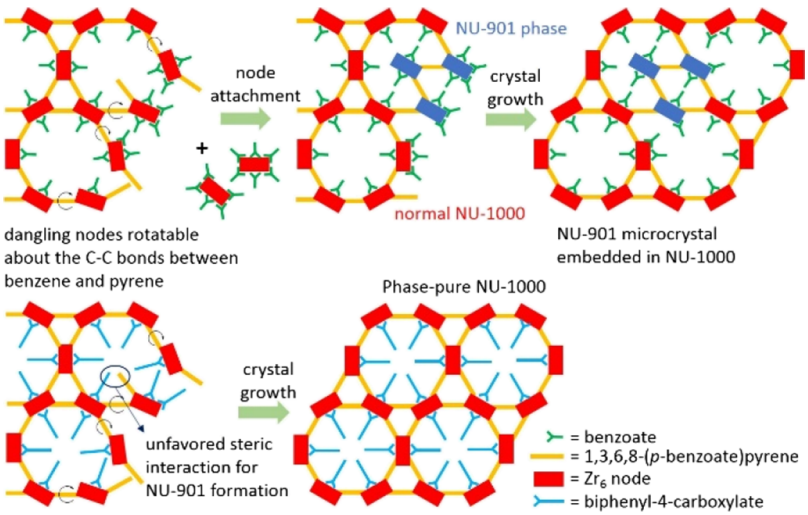


Figure 3. Schematic of the formation pathway that leads to NU-901-like domains within NU-1000 crystals and how substitution of benzoic acid with biphenyl-4-carboxylic acid as the modulator can lead to phase-pure NU-1000.

Scheme 1. Schematic Representation of the Modulator Exchange Reaction

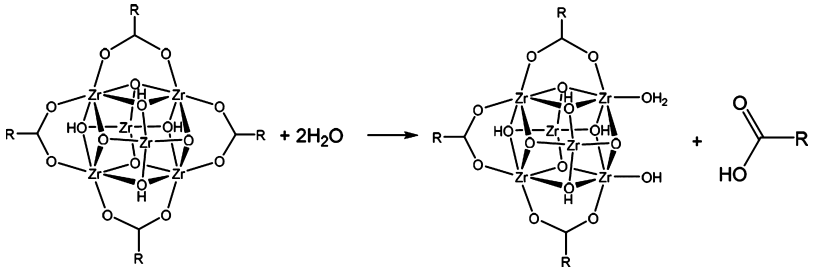


Table 1. Reaction Energies and Reaction-Free Energies for the Modulator Exchange Reaction Calculated at 378 K in DMF

R	H	CH_3	phenyl	biphenyl
ΔE	−5.9	−6.3	−7.0	−1.5
ΔG	8.0	7.2	3.3	NA

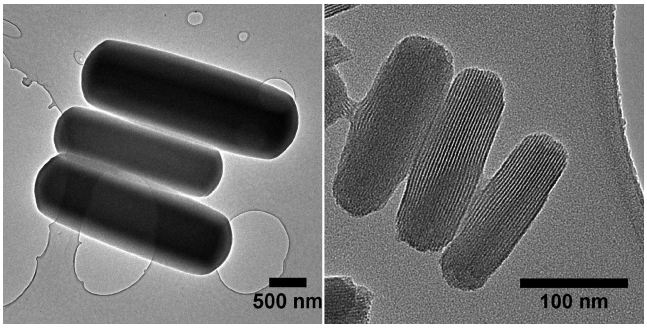


Figure 4. (Left) TEM image of NU-1000 crystals synthesized using the biphenyl-4-carboxylic acid modulator. The image shows homogeneous contrast through the length of the particle. (Right) Higher magnification TEM image of smaller NU-1000 crystals synthesized using the biphenyl-4-carboxylic acid modulator, in which the lattice fringe spacings are consistent with NU-1000 and span the entire lengths of the particles.

crystals are phase-pure NU-1000 and that the NU-901-like material is absent.

Finally, ^{1}H nuclear magnetic resonance spectroscopy (Figure S3) demonstrates the near-complete removal of the biphenyl-4-carboxylate modulator by way of the standard washing protocol.[14] This is an important consideration as a residual modulator could be expected to block the installation of metal precursors by atomic layer deposition, such as those employed by Kung et al.[29] and Kim et al.,[30] or bimetallic complexes by solution deposition, such as those employed by Thompson et al. and Desai et al.[22,23] In addition, the presence of modulator molecules could block metal exchange reactions as employed by Klet et al.[17] and block infiltration of precursors for silica nanocasting, as has been reported by Malonzo et al.[31]

To characterize the structurally homogeneous particles from the perspective of their potential to serve as scaffolds for anchored reactive sites, product materials were infiltrated with the dicobalt complex ((*N*,*N*,*N*-tris(2-(2-pyridylamino)ethyl)-amine)Co_2OPh[23]) via solution deposition, following the procedure detailed in the Supporting Information, using both structurally homogeneous and heterogeneous NU-1000 samples. Scanning TEM coupled with EDS results demonstrate a marked difference in cobalt distribution. In the NU-1000 synthesized using biphenyl-4-carboxylic acid, the distribution of cobalt throughout the crystals is homogeneous, whereas in the cobalt distribution in crystals synthesized using benzoic acid, a depletion zone is observed where the higher density material resides (Figure 5). In both cases, we observed loadings of one-to-two dicobalt complexes per Zr_6 node based on EDS quantification.

DOI: 10.1021/acsami.7b11348
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX

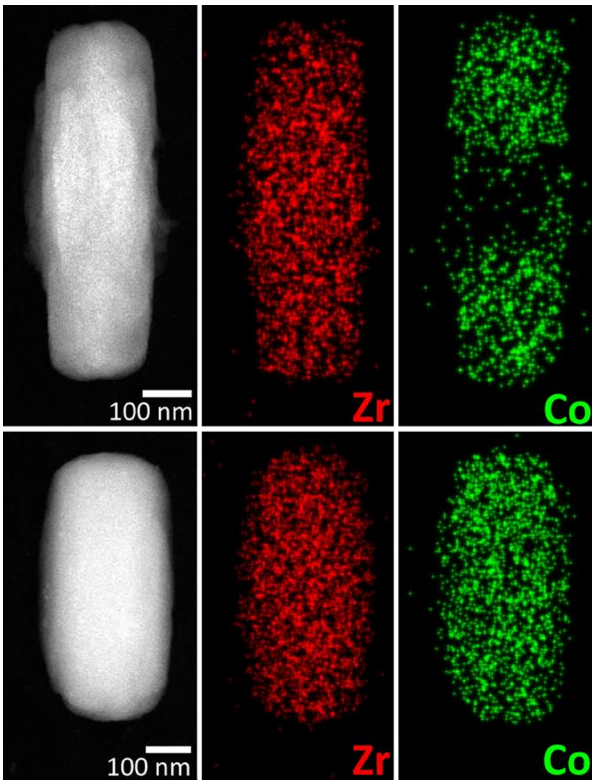


Figure 5. Representative high-angle annular dark-field images and energy-dispersive X-ray spectroscopy (EDS) maps of NU-1000 particles after solution deposition using the dicobalt complex: the material imaged in the top series was synthesized using benzoic acid and in the bottom series using biphenyl-4-carboxylic acid. The cobalt is uniformly distributed in the upper series, in stark contrast to the heterogeneous Co distribution observed in the top series, in which the NU-901-like phase is present.

CONCLUSIONS

The above results support the hypothesis that the bulkier modulator inhibits the arrangement of nodes that lead to the incorporation of the more dense material. Further, the increase in phase purity means that a higher fraction of nodes contained in the product crystal can be used for anchoring reactive metals. These results elucidate the role of the modulator in facilitating the incorporation of incoming species into the growing MOF crystal surface and are potentially generalizable to the wide field of modulated synthesis of MOFs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.7b11348.

Experimental procedures, characterization details, powder X-ray diffraction pattern, N_2 adsorption isotherm, and proton nuclear magnetic resonance (PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: rleepenn@umn.edu.

ORCID

Thomas E. Webber: 0000-0001-5728-2449
Wei-Guang Liu: 0000-0002-6633-7795
Connie C. Lu: 0000-0002-5162-9250
Donald G. Truhlar: 0000-0002-7742-7294
R. Lee Penn: 0000-0002-9610-9507

Author Contributions

T.E.W. and W.-G.L. contributed equally.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank Peter Dunn (University of Minnesota) for his assistance with ^{1}H NMR and Dr. Jason Myers (University of Minnesota) for his help and expertise with transmission electron microscopy. This work is supported as part of the Inorganometallic Catalyst Design Center, an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, Basic Energy Sciences under Award DE-SC0012702. Parts of this work were carried out in the Characterization Facility, University of Minnesota, which receives partial support from NSF through the MRSEC program.

REFERENCES

(1) Howarth, A. J.; Liu, Y.; Li, P.; Li, Z.; Wang, T. C.; Hupp, J. T.; Farha, O. K. Chemical, Thermal and Mechanical Stabilities of Metal–Organic Frameworks. *Nat. Rev. Mater.* **2016**, *1*, 15018.
(2) Rimoldi, M.; Howarth, A. J.; DeStefano, M. R.; Lin, L.; Goswami, S.; Li, P.; Hupp, J. T.; Farha, O. K. Catalytic Zirconium/Hafnium-Based Metal–Organic Frameworks. *ACS Catal.* **2017**, *7*, 997–1014.
(3) Faccioli, F.; Bauer, M.; Pedron, D.; Sorarù, A.; Carraro, M.; Gross, S. Hydrolytic Stability and Hydrogen Peroxide Activation of Zirconium-Based Oxoclusters. *Eur. J. Inorg. Chem.* **2015**, *2015*, 210–225.
(4) Bon, V.; Senkovska, I.; Baburin, I. A.; Kaskel, S. Zr- and Hf-Based Metal–Organic Frameworks: Tracking Down the Polymorphism. *Cryst. Growth Des.* **2013**, *13*, 1231–1237.
(5) Wang, T. C.; Bury, W.; Gómez-Gualdrón, D. A.; Vermeulen, N. A.; Mondloch, J. E.; Deria, P.; Zhang, K.; Moghadam, P. Z.; Sarjeant, A. A.; Snurr, R. Q.; Stoddart, J. F.; Hupp, J. T.; Farha, O. K. Ultrahigh Surface Area Zirconium MOFs and Insights into the Applicability of the BET Theory. *J. Am. Chem. Soc.* **2015**, *137*, 3585–3591.
(6) Zheng, J.; Wu, M.; Jiang, F.; Su, W.; Hong, M. Stable Porphyrin Zr and Hf Metal–Organic Frameworks Featuring 2.5 nm Cages: High Surface Areas, SCSC Transformations and Catalyses. *Chem. Sci.* **2015**, *6*, 3466–3470.
(7) Jiang, H.-L.; Feng, D.; Liu, T.-F.; Li, J.-R.; Zhou, H.-C. Pore Surface Engineering with Controlled Loadings of Functional Groups via Click Chemistry in Highly Stable Metal–Organic Frameworks. *J. Am. Chem. Soc.* **2012**, *134*, 14690–14693.
(8) Guillerm, V.; Ragon, F.; Dan-Hardi, M.; Devic, T.; Vishnuvarthan, M.; Campo, B.; Vimont, A.; Clet, G.; Yang, Q.; Maurin, G.; Férey, G.; Vittadini, A.; Gross, S.; Serre, C. A Series of Isoreticular, Highly Stable, Porous Zirconium Oxide Based Metal-Organic Frameworks. *Angew. Chem., Int. Ed.* **2012**, *51*, 9267–9271.
(9) Feng, D.; Gu, Z.-Y.; Chen, Y.-P.; Park, J.; Wei, Z.; Sun, Y.; Bosch, M.; Yuan, S.; Zhou, H.-C. A Highly Stable Porphyrinic Zirconium Metal–Organic Framework with shp-a Topology. *J. Am. Chem. Soc.* **2014**, *136*, 17714–17717.
(10) Feng, D.; Gu, Z.-Y.; Li, J.-R.; Jiang, H.-L.; Wei, Z.; Zhou, H.-C. Zirconium-Metalloporphyrin PCN-222: Mesoporous Metal-Organic Frameworks with Ultrahigh Stability as Biomimetic Catalysts. *Angew. Chem., Int. Ed.* **2012**, *51*, 10307–10310.
(11) Jiang, H.-L.; Feng, D.; Wang, K.; Gu, Z.-Y.; Wei, Z.; Chen, Y.-P.; Zhou, H.-C. An Exceptionally Stable, Porphyrinic Zr Metal–Organic Framework Exhibiting pH-Dependent Fluorescence. *J. Am. Chem. Soc.* **2013**, *135*, 13934–13938.
(12) Feng, D.; Chung, W.-C.; Wei, Z.; Gu, Z.-Y.; Jiang, H.-L.; Chen, Y.-P.; Darensbourg, D. J.; Zhou, H.-C. Construction of Ultrastable

DOI: 10.1021/acsami.7b11348
ACS Appl. Mater. Interfaces XXXX, XXX, XXX–XXX

Porphyrin Zr Metal−Organic Frameworks through Linker Elimination. *J. Am. Chem. Soc.* **2013**, *135*, 17105−17110.
(13) Furukawa, H.; Gándara, F.; Zhang, Y.-B.; Jiang, J.; Queen, W. L.; Hudson, M. R.; Yaghi, O. M. Water Adsorption in Porous Metal−Organic Frameworks and Related Materials. *J. Am. Chem. Soc.* **2014**, *136*, 4369−4381.
(14) Mondloch, J. E.; Bury, W.; Fairen-Jimenez, D.; Kwon, S.; Demarco, E. J.; Weston, M. H.; Sarjeant, A. A.; Nguyen, S. T.; Stair, P. C.; Snurr, R. Q.; Farha, O. K.; Hupp, J. T. Vapor-Phase Metalation by Atomic Layer Deposition in a Metal-Organic Framework. *J. Am. Chem. Soc.* **2013**, *135*, 10294−10297.
(15) Kung, C.-W.; Wang, T. C.; Mondloch, J. E.; Fairen-Jimenez, D.; Gardner, D. M.; Bury, W.; Klingsporn, J. M.; Barnes, J. C.; Van Duyne, R.; Stoddart, J. F.; Wasielewski, M. R.; Farha, O. K.; Hupp, J. T. Metal-Organic Framework Thin Films Composed of Free-Standing Acicular Nanorods Exhibiting Reversible Electrochromism. *Chem. Mater.* **2013**, *25*, 5012−5017.
(16) Peters, A. W.; Li, Z.; Farha, O. K.; Hupp, J. T. Atomically Precise Growth of Catalytically Active Cobalt Sulfide on Flat Surfaces and within a Metal−Organic Framework via Atomic Layer Deposition. *ACS Nano* **2015**, *9*, 8484−8490.
(17) Klet, R. C.; Wang, T. C.; Fernandez, L. E.; Truhlar, D. G.; Hupp, J. T.; Farha, O. K. Synthetic Access to Atomically Dispersed Metals in Metal−Organic Frameworks via a Combined Atomic-Layer-Deposition-in-MOF and Metal-Exchange Approach. *Chem. Mater.* **2016**, *28*, 1213−1219.
(18) Li, Z.; Schweitzer, N. M.; League, A. B.; Bernales, V.; Peters, A. W.; Getsoian, A. B.; Wang, T. C.; Miller, J. T.; Vjunov, A.; Fulton, J. L.; Lercher, J. A.; Cramer, C. J.; Gagliardi, L.; Hupp, J. T.; Farha, O. K. Sintering-Resistant Single-Site Nickel Catalyst Supported by Metal-Organic Framework. *J. Am. Chem. Soc.* **2016**, *138*, 1977−1982.
(19) Rimoldi, M.; Nakamura, A.; Vermeulen, N. A.; Henkelis, J. J.; Blackburn, A. K.; Hupp, J. T.; Stoddart, J. F.; Farha, O. K. A Metal−Organic Framework Immobilised Iridium Pincer Complex. *Chem. Sci.* **2016**, *7*, 4980−4984.
(20) Rimoldi, M.; Bernales, V.; Borycz, J.; Vjunov, A.; Gallington, L. C.; Platero-Prats, A. E.; Kim, I. S.; Fulton, J. L.; Martinson, A. B. F.; Lercher, J. A.; Chapman, K. W.; Cramer, C. J.; Gagliardi, L.; Hupp, J. T.; Farha, O. K. Atomic Layer Deposition in a Metal−Organic Framework: Synthesis, Characterization, and Performance of a Solid Acid. *Chem. Mater.* **2017**, *29*, 1058−1068.
(21) Platero-Prats, A. E.; Li, Z.; Gallington, L. C.; Peters, A. W.; Hupp, J. T.; Farha, O. K.; Chapman, K. W. Addressing the Characterisation Challenge to Understand Catalysis in MOFs: The Case of Nanoscale Cu Supported in NU-1000. *Faraday Discuss.* **2017**, *201*, 337−350.
(22) Thompson, A. B.; Pahls, D. R.; Bernales, V.; Gallington, L. C.; Malonzo, C. D.; Webber, T.; Tereniak, S. J.; Wang, T. C.; Desai, S. P.; Li, Z.; Kim, I. S.; Gagliardi, L.; Penn, R. L.; Chapman, K. W.; Stein, A.; Farha, O. K.; Hupp, J. T.; Martinson, A. B. F.; Lu, C. C. Installing Heterobimetallic Cobalt-Aluminum Single Sites on a Metal Organic Framework Support. *Chem. Mater.* **2016**, *28*, 6753−6762.
(23) Desai, S. P.; Malonzo, C.; Webber, T.; Duan, J.; Thompson, A. B.; Tereniak, S. J.; DeStefano, M. R.; Buru, C. T.; Li, Z.; Penn, R. L.; Farha, O.; Hupp, J. T.; Stein, A.; Lu, C. Assembly of Dicobalt and Cobalt-Aluminum Oxide Clusters on Metal−Organic Framework and Nanocast Silica Supports. *Faraday Discuss.* **2017**, *201*, 287−302.
(24) Wang, T. C.; Vermeulen, N. A.; Kim, I. S.; Martinson, A. B. F.; Stoddart, J. F.; Hupp, J. T.; Farha, O. K. Scalable Synthesis and Post-Modification of a Mesoporous Metal-Organic Framework Called NU-1000. *Nat. Protoc.* **2015**, *11*, 149−162.
(25) Hu, Z.; Castano, I.; Wang, S.; Wang, Y.; Peng, Y.; Qian, Y.; Chi, C.; Wang, X.; Zhao, D. Modulator Effects on the Water-Based Synthesis of Zr/Hf Metal−Organic Frameworks: Quantitative Relationship Studies between Modulator, Synthetic Condition, and Performance. *Cryst. Growth Des.* **2016**, *16*, 2295−2301.
(26) Schaate, A.; Roy, P.; Godt, A.; Lippke, J.; Waltz, F.; Wiebcke, M.; Behrens, P. Modulated Synthesis of Zr-Based Metal-Organic Frameworks: From Nano to Single Crystals. *Chem.—Eur. J.* **2011**, *17*, 6643−6651.
(27) Hu, Z.; Peng, Y.; Kang, Z.; Qian, Y.; Zhao, D. A Modulated Hydrothermal (MHT) Approach for the Facile Synthesis of UiO-66-Type MOFs. *Inorg. Chem.* **2015**, *54*, 4862−4868.
(28) Liu, W.-G.; Truhlar, D. G. Computational Linker Design for Highly Crystalline Metal-Organic Framework NU-1000. *Chem. Mater.* **2017**, *29*, 8073−8081.10.1021/acs.chemmater.7b01624
(29) Kung, C.-W.; Mondloch, J. E.; Wang, T. C.; Bury, W.; Hoffeditz, W.; Klahr, B. M.; Klet, R. C.; Pellin, M. J.; Farha, O. K.; Hupp, J. T. Metal−Organic Framework Thin Films as Platforms for Atomic Layer Deposition of Cobalt Ions to Enable Electrocatalytic Water Oxidation. *ACS Appl. Mater. Interfaces* **2015**, *7*, 28223−28230.
(30) Kim, I. S.; Borycz, J.; Platero-Prats, A. E.; Tussupbayev, S.; Wang, T. C.; Farha, O. K.; Hupp, J. T.; Gagliardi, L.; Chapman, K. W.; Cramer, C. J.; Martinson, A. B. F. Targeted Single-Site MOF Node Modification: Trivalent Metal Loading via Atomic Layer Deposition. *Chem. Mater.* **2015**, *27*, 4772−4778.
(31) Malonzo, C. D.; Shaker, S. M.; Ren, L.; Prinslow, S. D.; Platero-Prats, A. E.; Gallington, L. C.; Borycz, J.; Thompson, A. B.; Wang, T. C.; Farha, O. K.; Hupp, J. T.; Lu, C. C.; Chapman, K. W.; Myers, J. C.; Penn, R. L.; Gagliardi, L.; Tsapatsis, M.; Stein, A. Thermal Stabilization of Metal−Organic Framework-Derived Single-Site Catalytic Clusters through Nanocasting. *J. Am. Chem. Soc.* **2016**, *138*, 2739−2748.

DOI: 10.1021/acsami.7b11348
ACS Appl. Mater. Interfaces XXXX, XXX, XXX−XXX